Ordinary Norm-One Idempotent Multipliers of a Discrete Fourier Algebra Are Coset Indicators

Candidate full answer to the question in arXiv:0806.4643

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Status. Candidate full proof, likely valid, pending human review. The source question has a negative answer, and the proof gives a classification for every discrete group.

1 Source question and answer

B. E. Forrest and V. Runde, *Norm one idempotent cb-multipliers with applications to the Fourier algebra in the cb-multiplier norm*, arXiv:0806.4643 [1], characterize norm-one idempotents in $M_{\mathrm{cb}}A(G)$. In the remark following Theorem 1.1 (PDF page 4), they note that there are sets E in the free group $\mathbb{F}_2$ for which

$$\chi_E \in MA(\mathbb{F}_2) \setminus M_{\mathrm{cb}}A(\mathbb{F}_2),$$

and ask whether such an E can be chosen with ordinary multiplier norm $\|\chi_E\|_M = 1$.

> *Remark.* Let $MA(G)$ denote the algebra of all multipliers of $A(G)$. Defining $\|\phi\|_M$ as the operator norm of M_ϕ, we obtain a Banach algebra norm on $MA(G)$; obviously, $M_{cb}A(G)$ embeds contractively into $MA(G)$. Hence, every norm one idempotent in $M_{cb}A(G)$ is a norm one idempotent in $MA(G)$. By [Boż], $M_{cb}A(\mathbb{F}_2) \subsetneq MA(\mathbb{F}_2)$ holds, and—as M. Bożejko communicated to the second named author—, there are sets $E \subset \mathbb{F}_2$ such that $\chi_E \subset MA(\mathbb{F}_2) \setminus M_{cb}A(\mathbb{F}_2)$. We do not know if such E can be chosen such that $\|\chi_E\|_M = 1$.

Figure 1: The open question in the source PDF, page 4.

The answer is no. The following result is stronger because it applies to every discrete group.

Theorem 1. *Let G be a discrete group and let $\varnothing \neq E \subset G$. The following are equivalent:*

(i) $\chi_E \in MA(G)$ *and* $\|\chi_E\|_M = 1$*;*

(ii) *E is a left coset of a subgroup of G;*

(iii) $\chi_E \in M_{\mathrm{cb}}A(G)$ *and* $\|\chi_E\|_{M_{\mathrm{cb}}} = 1$.

Consequently no set requested in the source question exists.

2 Proof intuition

Translate E so that it contains the identity. On the dual $A(G)^* = VN(G)$, its multiplier becomes a norm-one projection which fixes the identity. That makes the projection positive. The discrete group von Neumann algebra has a faithful finite trace, and the Fourier-diagonal projection preserves it. Kadison's inequality then has no room to be strict on the range: every self-adjoint range element has its square in the range. Thus the range is closed under the ambient Jordan product.

The range is spanned weak-* by precisely the group unitaries indexed by the translated set. Jordan multiplying two such unitaries produces $\lambda(st) + \lambda(ts)$. Since group unitaries are linearly independent, both products must still be supported. Together with adjoint closure, this turns the translated set into a subgroup. Undoing the translation gives a coset, whose indicator is visibly a coefficient of a quasi-regular unitary representation.

3 Operator-algebra lemmas

We use the canonical faithful normal trace τ on $VN(G)$, determined by $\tau(\lambda(s)) = 0$ for $s \neq e$ and $\tau(1) = 1$.

Lemma 1. *If $T : \mathcal{A} \to \mathcal{B}$ is a unital linear contraction between unital C*-algebras, then T is positive.*

Proof. For every state ω on $\mathcal{B}$, the functional $\omega \circ T$ is unital and has norm at most one. Its norm is in fact one, and a unital norm-one functional on a C*-algebra is positive. Thus $\omega(T(a)) \geq 0$ whenever $a \geq 0$. States separate the positive cone of $\mathcal{B}$, so $T(a) \geq 0$. □

Lemma 2. *Let $\mathcal{M}$ be a finite von Neumann algebra with faithful normal trace τ, and let $P : \mathcal{M} \to \mathcal{M}$ be a unital positive projection such that $\tau \circ P = \tau$. Then $\operatorname{Ran} P$ is self-adjoint and is closed under the ambient Jordan product*

$$u \circ v = \frac{uv + vu}{2}.$$

Proof. A positive linear map preserves adjoints, so $\operatorname{Ran} P$ is self-adjoint. If $a = a^* \in \operatorname{Ran} P$, Kadison's inequality for unital positive maps [2] gives

$$P(a^2) \geq P(a)^2 = a^2.$$

The positive difference $P(a^2) - a^2$ has trace zero because P preserves τ. Faithfulness therefore forces $P(a^2) = a^2$, so $a^2 \in \operatorname{Ran} P$. For self-adjoint $a, b \in \operatorname{Ran} P$,

$$ab + ba = (a + b)^2 - a^2 - b^2 \in \operatorname{Ran} P.$$

Every range element is a complex linear combination of two self-adjoint range elements, and bilinearity finishes the proof. □

4 Proof of the classification

Proof of Theorem 1. Assume (i). Choose $x \in E$ and put $H = x^{-1}E$, so $e \in H$. Left translation of functions is an isometry of $A(G)$, and conjugating a multiplication operator by translations shows that

$$\chi_H \in MA(G), \qquad \|\chi_H\|_M = 1.$$

Let

$$P = M^*_{\chi_H} : VN(G) \longrightarrow VN(G).$$

For each $s \in G$, duality with point evaluation gives

$$P(\lambda(s)) = \chi_H(s)\lambda(s). \tag{1}$$

The map P is normal, $P^2 = P$, $\|P\| = 1$, and $P(1) = 1$ because $e \in H$. Lemma 1 makes P positive. In particular P preserves adjoints. If $s \in H$, then

$$\chi_H(s^{-1})\lambda(s^{-1}) = P(\lambda(s)^*) = P(\lambda(s))^* = \lambda(s^{-1}),$$

so $s^{-1} \in H$.

Equation (1) also shows on the group algebra that $\tau(P(a)) = \tau(a)$, since $\chi_H(e) = 1$. Both functionals are normal and the group algebra is weak-* dense in $VN(G)$, hence $\tau \circ P = \tau$ on all of $VN(G)$. Lemma 2 now applies.

Let $s, t \in H$. Then $\lambda(s), \lambda(t) \in \operatorname{Ran} P$, and Jordan closure gives

$$z := \lambda(st) + \lambda(ts) \in \operatorname{Ran} P.$$

Therefore $P(z) = z$. If $st = ts$, equation (1) immediately gives $\chi_H(st) = 1$. If $st \neq ts$, it gives

$$\chi_H(st)\lambda(st) + \chi_H(ts)\lambda(ts) = \lambda(st) + \lambda(ts).$$

Distinct group unitaries are linearly independent (apply them to $\delta_e \in \ell^2(G)$), so again $\chi_H(st) = 1$; indeed both coefficients are one. Thus H contains e and is closed under products and inverses. It is a subgroup, and $E = xH$. This proves (i)⇒(ii).

For (ii)⇒(iii), let $E = xH$ and let π be the quasi-regular representation on $\ell^2(G/H)$. Then

$$\chi_{xH}(s) = \langle \pi(s)\delta_H, \delta_{xH} \rangle.$$

Hence $\chi_E \in B(G)$ with $B(G)$-norm at most one. Since it takes the value one, the norm is exactly one. The canonical contractive inclusion $B(G) \hookrightarrow M_{\mathrm{cb}}A(G)$ gives cb multiplier norm at most one, and a nonzero idempotent operator has norm at least one. Thus the cb norm is one.

Finally, the contractive inclusion $M_{\mathrm{cb}}A(G) \hookrightarrow MA(G)$ gives (iii)⇒(i), again using that a nonzero idempotent has norm at least one. □

5 Verification, scope, and novelty

The proof uses ordinary multiplier contractivity only; complete boundedness enters after the support has already been classified as a coset. Discreteness is used exactly in the availability of the canonical faithful finite trace on $VN(G)$. Therefore this packet does not assert the analogous classification for nondiscrete locally compact groups, and it does not address the distinct amenability questions later in the source paper.

As a finite check, the exact abelian Fourier formula was used to enumerate all 8,177 nonempty subsets of cyclic groups C_n, $2 \leq n \leq 12$. Ordinary multiplier norm one occurred exactly for cosets. The code is included in `code/check_cyclic_groups.py`; the proof does not depend on it.

The run's lightweight indexes were searched for arXiv id 0806.4643 and the core ordinary-multiplier/coset terms. A bounded external search on 11 August 2026 checked the exact source wording, close norm-one idempotent variants, the source's citation graph, and metadata for the closest later work on idempotent multiplier norms and contractive projections. No exact answer or statement of the all-discrete-group classification was found. This was not an exhaustive literature review; originality is therefore a candidate claim pending expert checking.

The recommended human-review focus is the short trace-to-Jordan argument and the Fourier-diagonal identification (1). No conditional dependency remains.

References

[1] B. E. Forrest and V. Runde, Norm one idempotent cb-multipliers with applications to the Fourier algebra in the cb-multiplier norm, arXiv:0806.4643 (2008), arXiv:0806.4643.

[2] R. V. Kadison, A generalized Schwarz inequality and algebraic invariants for operator algebras, *Ann. of Math.* (2) **56** (1952), 494–503.